BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Annual Examination 2026

Mathematics — SSC-I (9th Grade)

Syllabus: Chapter 3 — Sets and Relations

ANSWER KEY (Version 1)

Total Marks: 65	**Time Allowed: 2h 30m**	**Version: 1**

SECTION A — MCQ Answers (12 × 1 = 12 Marks)

Q1. Correct: **B — distinct objects** **1 mark**
A set is defined as a **well-defined collection of distinct objects**. "Similar" is wrong because a set can hold objects of different types (e.g., a star, a moon, a tube light). "Ordered" is wrong because listing order in a set does not matter. "Finite" is wrong because sets can be infinite (e.g., set of natural numbers).

Q2. Correct: **B — $\{x \mid x \in A \wedge x \notin B\}$** **1 mark**
The **difference $A - B$** is the collection of elements that belong to A but do NOT belong to B. Option A misses the "not in B" condition. Option C is the intersection. Option D would give elements only from B.

Q3. Correct: **A — $B \subset A$** **1 mark**
If $A \cup B = A$, every element of B is already in A, so **$B \subset A$**. This is confirmed by $A \cap B = B$ (every element of B is common to both). Students often confuse which set is the subset.

Q4. Correct: **C — $A \cap B = \varnothing$** **1 mark**
Two sets are **disjoint** when they share no common element, i.e., their intersection is the empty set $\varnothing$. $A \cup B = \varnothing$ would mean both sets are empty, which is a different (and very specific) situation.

Q5. Correct: **B — $(A \cup B) \cup C = A \cup (B \cup C)$** **1 mark**
The **associative property of union** states that the grouping of three sets under union does not affect the result. Option D is the commutative property. Option A mixes union and intersection incorrectly.

Q6. Correct: **A — $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$** **1 mark**
The **distributive law of union over intersection** says union distributes over intersection the same way multiplication distributes over addition. Option B replaces the outer $\cap$ with $\cup$, making it wrong.

Q7. Correct: **C — $n(A) + n(B) - n(A \cap B)$** **1 mark**
The inclusion-exclusion formula for **overlapping sets** subtracts $n(A \cap B)$ to avoid counting common elements twice. For disjoint sets the simpler $n(A) + n(B)$ applies, but that is a special case not valid here.

Q8. Correct: **C — p^2** **1 mark**
The key fact from the chapter: **$n(A \times A) = n(A) \times n(A) = p \times p = p^2$**. Students confuse this with $n(A \times B) = n(A) \times n(B)$ and mistakenly write $2p$ (addition rather than multiplication).

Q9. Correct: **C — $a = c$ and $b = d$** **1 mark**
Two ordered pairs (a, b) and (c, d) are equal if and only if **both first elements are equal and both second elements are equal**. Order matters: $(a, b) \neq (b, a)$ in general, so option A (swapped condition) is wrong.

Q10. Correct: **D — 2^{t^2}** **1 mark**
$n(B) = t$, so $n(B \times B) = t^2$. The number of binary relations in $B \times B$ equals the number of subsets of $B \times B$, which is **2^{t^2}**. This is a direct application of the formula: number of binary relations in $A \times B = 2^{n(A\times B)}$.

Q11. Correct: **B — Domain** **1 mark**
The **domain** of a binary relation is the set of all first elements of its ordered pairs. The range is the set of all second elements. Students commonly swap domain and range.

Q12. Correct: **B — $\{2, 4\}$** **1 mark**
$R = \{(2,1), (4,3), (2,2)\}$ so the first elements are 2, 4, 2. Since a **set does not contain repeated elements**, Dom $R = \{2, 4\}$. Option A incorrectly lists 2 twice. Option C lists the second elements (range), not the domain.

SECTION B — Short Question Answers ($11 \times 3 = 33$ Marks)

Q 2(i) — Set Builder Form

B = Set of prime numbers less than 17

Primes less than 17: 2, 3, 5, 7, 11, 13 **1 mark**

Set builder form: $B = \{x \mid x \in P \wedge x < 17\}$ **1 mark**

where P denotes the set of prime numbers. **1 mark**

OR

$D = \{1, 2, 3, 6\}$: $\mathbf{D = \{x \mid x \text{ is a factor of } 6\}}$ **1 mark**

$G = \{15x \mid x \in Z \wedge x \geq 1\}$ in tabular form:

When $x = 1$: 15; $x = 2$: 30; $x = 3$: 45; ...

$\mathbf{G = \{15, 30, 45, 60, \ldots\}}$ (set of positive integral multiples of 15) **2 marks**

Q 2(ii) — Union and Intersection

$A = \{1, 2, 3, 4, 5\}$, $B = \{3, 4, 1, 7, 6\}$

$\mathbf{A \cup B}$ = all elements from A or B = $\{1, 2, 3, 4, 5, 6, 7\}$ **1 mark**

$\mathbf{A \cap B}$ = elements common to both = $\{1, 3, 4\}$ **1 mark**

(Common elements are written only once in the union.) **1 mark**

OR

$U = \{1, 2, 3, 4, 5, 6, 7\}$, $A = \{2, 4, 6\}$

$\mathbf{A^c = U - A}$ = $\{1, 3, 5, 7\}$ **1 mark**

Verification: $A \cup A^c = \{2, 4, 6\} \cup \{1, 3, 5, 7\} = \{1, 2, 3, 4, 5, 6, 7\} = U$ √ **1 mark**

$A \cap A^c = \{2, 4, 6\} \cap \{1, 3, 5, 7\} = \{\} = \varnothing$ √ **1 mark**

Q 2(iii) — Difference of Sets

$A = \{1, 2, 3, 4, 5\}$, $B = \{3, 4, 6, 7, 8\}$

$\mathbf{A - B}$ = elements in A not in B = $\{1, 2, 5\}$ **1 mark**

$\mathbf{B - A}$ = elements in B not in A = $\{6, 7, 8\}$ **1 mark**

Since $\{1, 2, 5\} \neq \{6, 7, 8\}$, therefore $\mathbf{A - B \neq B - A}$ (in general, set difference is not commutative). **1 mark**

OR

Disjoint sets: Two sets A and B with $A \cap B = \varnothing$ (no common element). Example: $\{1, 2, 3\}$ and $\{4, 5\}$ are disjoint. **1 mark**

Venn diagram: two separate non-overlapping circles inside a rectangle. **1 mark**

Overlapping sets: Two sets where (i) at least one element is common and (ii) neither is a subset of the other. Example: $\{2, 3, 5, 7, 9\}$ and $\{0, 3, 5, 8\}$. **1 mark**

Q 2(iv) — Three-set Union (Venn diagram)

$X = \{a, b, c, d, e\}$, $Y = \{a, c, e\}$, $Z = \{g, h, i, j\}$

Step 1: $X \cup Y = \{a, b, c, d, e\}$ (since $Y \subset X$) **1 mark**

Step 2: $(X \cup Y) \cup Z = \{a, b, c, d, e\} \cup \{g, h, i, j\} = \{a, b, c, d, e, g, h, i, j\}$ **1 mark**

Venn diagram shows three circles; X and Y overlap (Y inside X); Z is separate from X and Y. The entire shaded region is the answer above. **1 mark**

OR

$P = \{0, 1, 2, 3\}$, $Q = \{2, 3, 4, 5, 6\}$, $R = \{5, 6, 7, 8, 9\}$

Step 1: $Y \cup Z$ (i.e., $Q \cup R$) $= \{2, 3, 4, 5, 6, 7, 8, 9\}$ **1 mark**

Step 2: $P \cap (Q \cup R) = \{0,1,2,3\} \cap \{2,3,4,5,6,7,8,9\} = \{2, 3\}$ **1 mark**

Venn diagram shows three circles with appropriate overlapping regions shaded. **1 mark**

Q 2(v) — Associative Property of Intersection

$A = \{1, 2, 3, 5, 7, 9\}$, $B = \{1, 2, 3, 4, 5, 6\}$, $C = \{5, 6, 7, 8\}$

LHS: $(A \cap B) \cap C$

$A \cap B = \{1, 2, 3, 5\}$ **1 mark**

$(A \cap B) \cap C = \{1,2,3,5\} \cap \{5,6,7,8\} = \{5\}$ **1 mark**

RHS: $A \cap (B \cap C)$

$B \cap C = \{5, 6\}$

$A \cap (B \cap C) = \{1,2,3,5,7,9\} \cap \{5,6\} = \{5\}$ **1 mark**

LHS = RHS = $\{5\}$ $\therefore$ Associative property of intersection is verified.

OR

$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

$B \cup C = \{1,2,3,4,5,6,7,8\}$; $A \cap (B \cup C) = \{1,2,3,5,7\}$ **1 mark**

$A \cap B = \{1,2,3,5\}$; $A \cap C = \{5,7\}$ **1 mark**

$(A \cap B) \cup (A \cap C) = \{1,2,3,5\} \cup \{5,7\} = \{1,2,3,5,7\}$ **1 mark**

LHS = RHS $\therefore$ Distributive property verified.

Q 2(vi) — Application: Class Survey (Two Sets)

Given: $n(\text{Total}) = 80$, $n(E) = 40$, $n(M) = 34$, $n(E \cap M) = 9$

$n(E \cup M) = n(E) + n(M) - n(E \cap M)$ $= 40 + 34 - 9 =$ **65** students like either or both subjects. **2 marks**

Students who like neither $= n(\text{Total}) - n(E \cup M) = 80 - 65 =$ **15** **1 mark**

OR

$n(A) = 24$, $n(B) = 18$, $n(A \cup B) = 31$

Using: $n(A \cup B) = n(A) + n(B) - n(A \cap B)$ **1 mark**

$31 = 24 + 18 - n(A \cap B)$ **1 mark**

$n(A \cap B) = 42 - 31 = 11$ **1 mark**

Q 2(vii) — Application: Houses Survey

$n(U) = 50$, $n(L) = 25$ (lawns), $n(C) = 32$ (car porches), $n(L \cap C) = 15$

$n(L \cup C) = 25 + 32 - 15 =$ **42** houses have either or both features. **2 marks**

Houses with **neither** $= 50 - 42 =$ **8** **1 mark**

OR

$n(U) = 70$, $n(T) = 48$ (tea), $n(C) = 40$ (coffee), every person likes at least one.

So $n(T \cup C) = 70$ **1 mark**

$n(T \cup C) = n(T) + n(C) - n(T \cap C)$

$70 = 48 + 40 - n(T \cap C)$ **1 mark**

$n(T \cap C) = 88 - 70 = 18$ people like both. **1 mark**

Q 2(viii) — Equality of Ordered Pairs

$(x - 3y, 5x + 1) = (4, 6)$

By equality of ordered pairs:

$5x + 1 = 6 \Rightarrow 5x = 5 \Rightarrow$ **$x = 1$** **1 mark**

$x - 3y = 4 \Rightarrow 1 - 3y = 4 \Rightarrow -3y = 3 \Rightarrow$ **$y = -1$** **2 marks**

OR

$(2a - 4, 6) = (8, -b + 1)$

$2a - 4 = 8 \Rightarrow 2a = 12 \Rightarrow$ **$a = 6$** **1 mark**

$-b + 1 = 6 \Rightarrow -b = 5 \Rightarrow$ **$b = -5$** **2 marks**

Q 2(ix) — Cartesian Product

$A = \{1, 4, 8\}$, $B = \{1, 0\}$

$A \times B$ $= \{(1,1), (1,0), (4,1), (4,0), (8,1), (8,0)\}$ — 6 elements **1 mark**

$B \times A$ $= \{(1,1), (1,4), (1,8), (0,1), (0,4), (0,8)\}$ — 6 elements **1 mark**

$n(A \times B) = n(B \times A) = n(A) \times n(B) = 3 \times 2 = 6$. Note $A \times B \neq B \times A$ (different pairs). **1 mark**

OR

$L \times M = \{(0,2),(0,3),(0,4),(1,2),(1,3),(1,4)\}$

First elements: 0, 1 ⇒ **$L = \{0, 1\}$** **1 mark**

Second elements: 2, 3, 4 ⇒ **$M = \{2, 3, 4\}$** **1 mark**

$M \times L = \{(2,0),(2,1),(3,0),(3,1),(4,0),(4,1)\}$ **1 mark**

Q 2(x) — Binary Relation and Arrow Diagram

$A = \{1, 2, 3\}$, $B = \{0, 1, 3\}$; $R = \{(a,b) \mid a \in A, b \in B, a > b\}$

Checking all pairs in $A \times B$: (1,0)√ (2,0)√ (2,1)√ (3,0)√ (3,1)√

$R = \{(1,0), (2,0), (2,1), (3,0), (3,1)\}$ **2 marks**

Arrow diagram: left oval {1,2,3}, right oval {0,1,3}, arrows from each pair above drawn with arrowheads. **1 mark**

OR

$C = \{2,4,6\}$, $D = \{4,6,8,9,12\}$; $R = \{(x,y) \mid x \text{ is factor of } y\}$

2 is factor of: 4, 6, 8, 12; 4 is factor of: 4, 8, 12; 6 is factor of: 6, 12

$R = \{(2,4),(2,6),(2,8),(2,12),(4,4),(4,8),(4,12),(6,6),(6,12)\}$ **1 mark**

Dom $R = \{2, 4, 6\}$ **1 mark**

Range $R = \{4, 6, 8, 12\}$ **1 mark**

Q 2(xi) — Inverse Relation

$A = \{0, 2, 3\}$, $B = \{0, 2, 4, 6, 9, 16\}$; $R = \{(x,y) \mid x^2 = y\}$

$0^2 = 0$ √, $2^2 = 4$ √, $3^2 = 9$ √

$R = \{(0,0), (2,4), (3,9)\}$ **1 mark**

Dom $R = \{0, 2, 3\}$; Range $R = \{0, 4, 9\}$ **1 mark**

$R^{-1} = \{(0,0), (4,2), (9,3)\}$; Dom $R^{-1} = \{0,4,9\}$ = Range R √ **1 mark**

OR

$R = \{(2,0),(4,2),(6,4),(8,6),(10,8)\}$

Pattern: second element = first element − 2, so set builder: **$R = \{(x,y) \mid x \in S, y = x - 2\}$** (where $S = \{2,4,6,8,10\}$) **1 mark**

Dom $R = \{2,4,6,8,10\}$; Range $R = \{0,2,4,6,8\}$ **1 mark**

$R^{-1} = \{(0,2),(2,4),(4,6),(6,8),(8,10)\}$ **1 mark**

SECTION C — Long Question Answers (4 × 5 = 20 Marks)

Q 3(i) — Distributive Laws (Venn Diagram Verification)

U = {0,1,2,3,4,5,6,7,8,9,10,11,12}, A = {0,1,2,3,4}, B = {3,4,5,6,7}, C = {3,4,5,6,7} (same as B here since both defined $3 \le y \le 7$)

(a) Verify A ∪ (B ∩ C) = (A ∪ B) ∩ (A ∪ C)

B ∩ C = {3,4,5,6,7} **1 mark**

LHS: A ∪ (B ∩ C) = {0,1,2,3,4} ∪ {3,4,5,6,7} = {0,1,2,3,4,5,6,7} **1 mark**

A ∪ B = {0,1,2,3,4,5,6,7}; A ∪ C = {0,1,2,3,4,5,6,7}

RHS: (A ∪ B) ∩ (A ∪ C) = {0,1,2,3,4,5,6,7} **1 mark**

LHS = RHS ∴ Distributive property of union over intersection is verified.

(b) Verify A ∩ (B ∪ C) = (A ∩ B) ∪ (A ∩ C)

B ∪ C = {3,4,5,6,7}; LHS: A ∩ {3,4,5,6,7} = {3,4} **1 mark**

A ∩ B = {3,4}; A ∩ C = {3,4}; RHS: {3,4} ∪ {3,4} = {3,4} **1 mark**

LHS = RHS ∴ Distributive property of intersection over union is verified.

OR

X = {2,4,6,8,10,12,14,16,18} (2x, x ∈ N, x < 10, giving 9 multiples); Y = {3,6,9,12,15,18} (first 6 multiples of 3); Z = {0,6,12,18} (6x, x ∈ W, x < 4 gives 0,6,12,18)

(a) (X ∪ Y) ∪ Z = X ∪ (Y ∪ Z)

X ∪ Y = {2,3,4,6,8,9,10,12,14,15,16,18} **1 mark**

(X ∪ Y) ∪ Z adds {0}: {0,2,3,4,6,8,9,10,12,14,15,16,18} **1 mark**

Y ∪ Z = {0,3,6,9,12,15,18}; X ∪ (Y ∪ Z) = same set above **1 mark**

(b) (X ∩ Y) ∩ Z = X ∩ (Y ∩ Z)

X ∩ Y = {6,12,18}; (X ∩ Y) ∩ Z = {6,12,18} **1 mark**

Y ∩ Z = {0,6,12,18} ∩ {3,6,9,12,15,18} = {6,12,18}; X ∩ (Y ∩ Z) = {6,12,18} **1 mark**

Both associative properties verified.

Q 3(ii) — Three-Set Survey: Channels

Given: n(U) = 60, n(A) = 25, n(B) = 16, n(C) = 13, n(A ∩ B) = 4, n(B ∩ C) = 7, n(A ∩ C) = 8, n(A ∩ B ∩ C) = 3

Formula: n(A ∪ B ∪ C) = n(A) + n(B) + n(C) − n(A ∩ B) − n(B ∩ C) − n(A ∩ C) + n(A ∩ B ∩ C) **2 marks**

= 25 + 16 + 13 − 4 − 7 − 8 + 3 **2 marks**

= 57 − 19 + 3 = **38** people watch at least one channel.

(ii) People who watch none = n(U) − n(A ∪ B ∪ C) = 60 − 38 = **22** **1 mark**

OR

n(U) = 100, n(M) = 45, n(Sc) = 40, n(H) = 50, n(M ∩ Sc) = 12, n(Sc ∩ H) = 15, n(H ∩ M) = 20, n(M ∩ Sc ∩ H) = 5

(i) Venn diagram: three overlapping circles inside rectangle, with values filled region by region:

Only M = 45 − 12 − 20 + 5 = 18; Only Sc = 40 − 12 − 15 + 5 = 18; Only H = 50 − 15 − 20 + 5 = 20

M ∩ Sc only = 7; Sc ∩ H only = 10; H ∩ M only = 15; centre = 5 **2 marks**

(ii) n(M ∪ Sc ∪ H) = 45 + 40 + 50 − 12 − 15 − 20 + 5 = 93 passed at least one. **2 marks**

(iii) Did not pass any = 100 − 93 = **7** **1 mark**

Q 3(iii) — Cartesian Product Distributive Law

A = {1,3,5}, B = {2,4}, C = {6,7}

(i) B ∪ C = {2,4,6,7}

A × (B ∪ C) = {(1,2),(1,4),(1,6),(1,7),(3,2),(3,4),(3,6),(3,7),(5,2),(5,4),(5,6),(5,7)} — 12 pairs **1 mark**

(ii) A × B = {(1,2),(1,4),(3,2),(3,4),(5,2),(5,4)}
A × C = {(1,6),(1,7),(3,6),(3,7),(5,6),(5,7)} **1 mark**

(A × B) ∪ (A × C) = {(1,2),(1,4),(3,2),(3,4),(5,2),(5,4),(1,6),(1,7),(3,6),(3,7),(5,6),(5,7)} — 12 pairs **1 mark**

(iii) A × (B ∪ C) = (A × B) ∪ (A × C) since both give the same 12 ordered pairs. Verified √ **1 mark**

(iv) n(A × B) = 3 × 2 = 6, so number of subsets = 2^6 = **64** **1 mark**

OR

D = {a,e,i}, E = {a,c}, F = {b,c}

(i) E ∩ F = {c}

D × (E ∩ F) = {(a,c),(e,c),(i,c)} **1 mark**

(ii) D × E = {(a,a),(a,c),(e,a),(e,c),(i,a),(i,c)}
D × F = {(a,b),(a,c),(e,b),(e,c),(i,b),(i,c)}
(D × E) ∩ (D × F) = {(a,c),(e,c),(i,c)} **1 mark**

(iii) Both give {(a,c),(e,c),(i,c)}, so D × (E ∩ F) = (D × E) ∩ (D × F). Verified √ **1 mark**

(iv) Arrow diagram for D × E: left oval {a,e,i}, right oval {a,c}, arrows from each d ∈ D to each e ∈ E. Arrow diagram for E × D: left oval {a,c}, right oval {a,e,i}. D × E ≠ E × D since D ≠ E (and pairs differ). **2 marks**

Q 3(iv) — Binary Relations: Domain, Range and Inverse

S = {1,2,4,8}, T = $\{3^0,3^1,3^2\}$ = {1,3,9}

(i) R_1 = {(x,y) | x = y}

Check: 1 ∈ S and 1 ∈ T? Yes. 3 ∈ S? No. 9 ∈ S? No.

R_1 = {(1,1)}; Dom R_1 = {1}; Range R_1 = {1} **1 mark**

(ii) R_2 = {(x,y) | y < x}

(2,1)√ (4,1)√ (4,3)√ (8,1)√ (8,3)√

R_2 = {(2,1),(4,1),(4,3),(8,1),(8,3)}; Dom R_2 = {2,4,8}; Range R_2 = {1,3} **1 mark**

(iii) R_3 = {(x,y) | x + y ∈ E} (E = even numbers)

x + y is even when both are odd or both are even. Odd in S: {1}; even in S: {2,4,8}. Odd in T: {1,3,9}; even in T: none.

(1,1): 2 even√ (1,3): 4 even√ (1,9): 10 even√ (2,?): 2+1=3 odd; 2+3=5 odd; 2+9=11 odd — none. Same for 4 and 8.

R_3 = {(1,1),(1,3),(1,9)}; Dom R_3 = {1}; Range R_3 = {1,3,9} **2 marks**

(iv) R_1^{-1} = {(1,1)}

Dom R_1^{-1} = {1} = Range R_1 √ **1 mark**

OR

A = {0,1,3}, B = {1,2,3,5,7}; R = {(x,y) | y = 2x + 1}

x=0: y=1√ x=1: y=3√ x=3: y=7√

R = {(0,1),(1,3),(3,7)}; Dom R = {0,1,3}; Range R = {1,3,7} **1 mark**

R^{-1} = {(1,0),(3,1),(7,3)}; Dom R^{-1} = {1,3,7}; Range R^{-1} = {0,1,3} **1 mark**

Arrow diagram for R: left oval {0,1,3}, right oval {1,2,3,5,7}, arrows 0→1, 1→3, 3→7. **1 mark**

Arrow diagram for R^{-1}: directions reversed: 1→0, 3→1, 7→3. **1 mark**

Verification: Dom R^{-1} = {1,3,7} = Range R √ and Range R^{-1} = {0,1,3} = Dom R √ **1 mark**

— End of Answer Key —